

Use the logic gate symbols

9608_s21_qp_13_sc_2.b

QUESTION

(b) Identify **four** logic gates used in the logic circuit above.

1

2

3

4 [1]

MARK SCHEME

2(b)	1 mark for all four: NOT NAND OR XOR	1
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9608_s21_qp_12_sc_1.b

QUESTION

(b) Identify the **three** logic gates used in the given logic circuit.

Gate 1

Gate 2

Gate 3 [1]

MARK SCHEME

1(b)	OR, AND, NOT	1
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QUESTION

(b) Identify **three** logic gates **not** used in the logic circuit.

Gate 1

Gate 2

Gate 3

[1]

MARK SCHEME

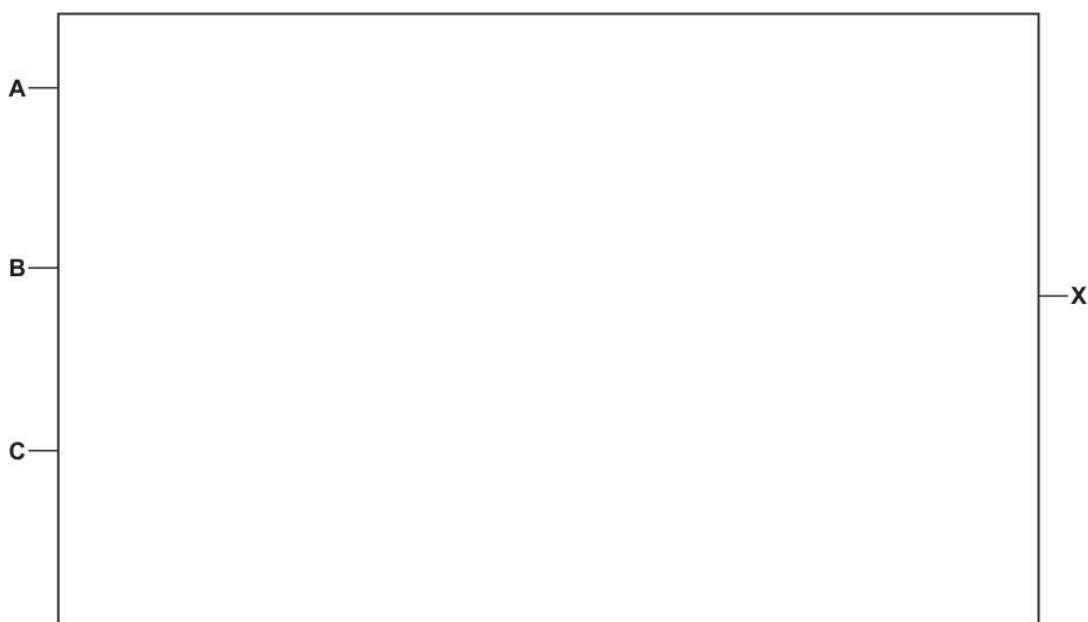
4(b)	1 mark for all three gates: OR XOR NOT	1
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QUESTION

3 (a) The following is a logic expression.

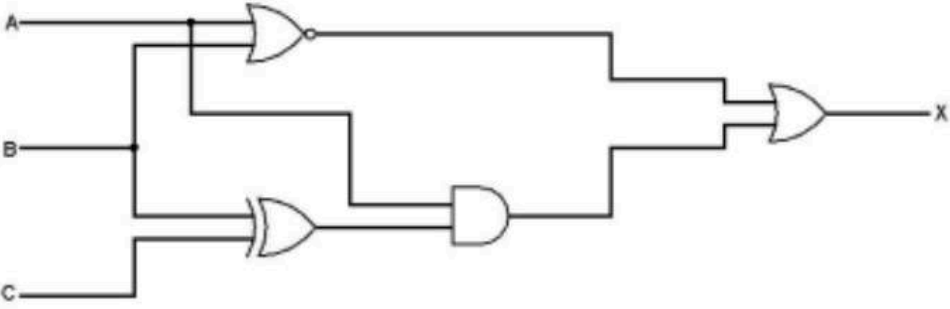
$$X = \text{NOT}(A \text{ OR } B) \text{ OR } (A \text{ AND } (B \text{ XOR } C))$$

Draw the logic circuit for the given expression, using a maximum of **four** logic gates.



[4]

MARK SCHEME

3(a)	1 mark per correct gate • A NOR B • B XOR C • A AND (B XOR C) • Final OR 	4
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9608_s20_qp_13_sc_3.c

QUESTION

(c) The following is a logic expression. A AND B XOR C OR NOT A Identify one logic gate that would not be used in the logic circuit for this expression. Draw the symbol for the logic gate. Logic gate Logic gate symbol: [2]
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MARK SCHEME

3(c)	1 mark for name, 1 mark for symbol matching the name NAND  NOR 	2
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QUESTION

- 3 The following is a logic expression.

$$X = \text{NOT } (A \text{ AND } B) \text{ OR NOT } (\text{NOT } B \text{ OR } C)$$

Draw the logic circuit for the given expression using a maximum of **four** gates.

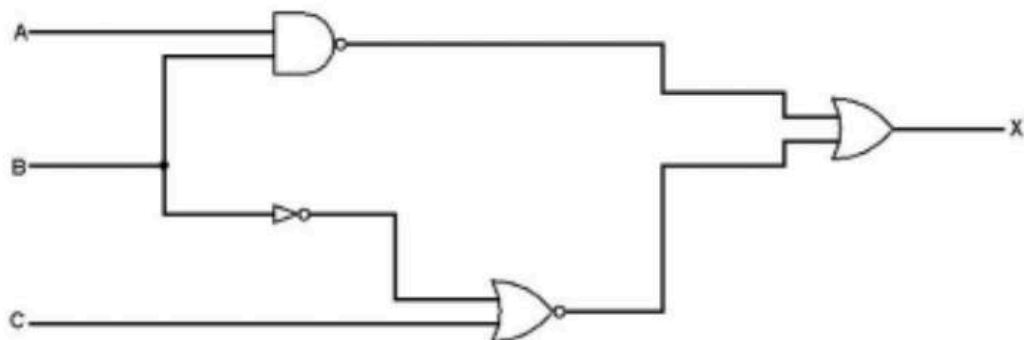


MARK SCHEME

3

1 mark for each correct gate

$$X = \text{NOT } (A \text{ AND } B) \text{ OR NOT } (\text{NOT } B \text{ OR } C)$$



4

QUESTION

(b) Identify **one** logic gate not used in the logic circuit in **part (a)**.

Draw the symbol for this logic gate **and** complete its truth table.

Logic gate:

Symbol:

Truth table:

Input		Output
A	B	
0	0	
0	1	
1	0	
1	1	

[3]

MARK SCHEME

5(b)

1 mark for name, **1 mark** for symbol, **1 mark** for truth table

3

- NAND



Input		Output
A	B	
0	0	1
0	1	1
1	0	1
1	1	0

- NOR



Input		Output
A	B	
0	0	1
0	1	0
1	0	0
1	1	0

QUESTION

- 5 (a) Draw a logic circuit to represent the logic expression:

$$X = \text{NOT } (A \text{ OR } C) \text{ OR } (A \text{ AND NOT } B)$$



[5]

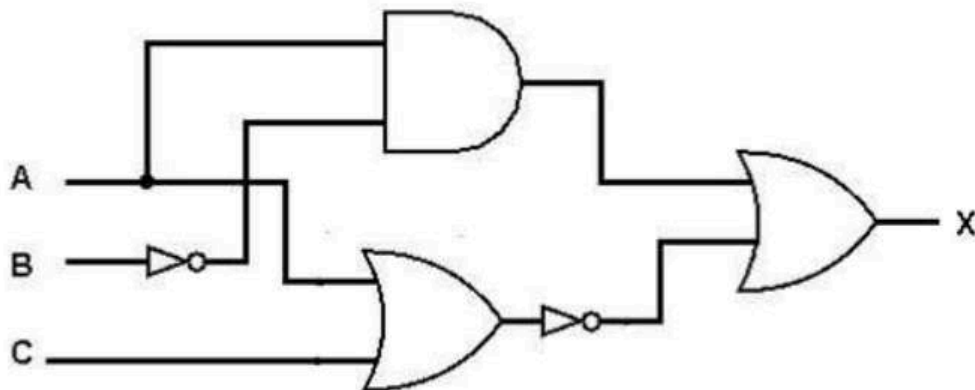
MARK SCHEME

5(a)

1 mark for each correct gate

5

A OR C
NOT(A OR C)
NOT B
A AND NOT B
Final OR



QUESTION

- 3 (a) A bank approves a customer for an account based on the criteria in the following table.

Parameter	Description of parameter	Binary value	Condition
A	Employed	1	True
		0	False
B	Self-employed	1	True
		0	False
C	Over 21	1	True
		0	False
D	Earn more than 30 000	1	True
		0	False
E	Another account	1	True
		0	False

A customer is approved ($X = 1$) if the person:

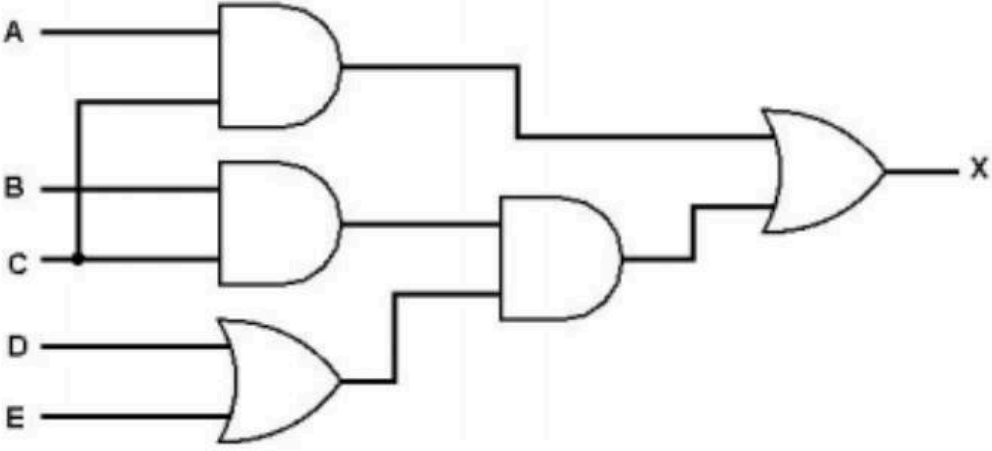
- is over 21 **and** employed
or
- is over 21 **and** self-employed **and**
 - **either** earns more than 30 000
or
 - has another account.

Draw a logic circuit to represent the model.



[5]

MARK SCHEME

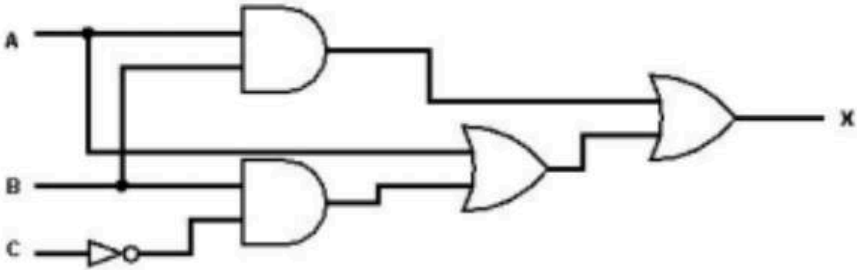
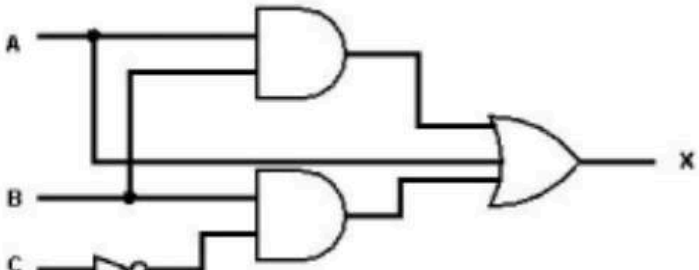
3(a)	1 mark for each gate A AND C B AND C D OR E (B AND C) AND (D OR E) Final OR 	5
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9608_s19_qp_12_sc_6.a

QUESTION

6 (a) Draw a logic circuit to represent the logic expression: $X = A \text{ OR } (B \text{ AND NOT } C) \text{ OR } (A \text{ AND } B)$  [5]

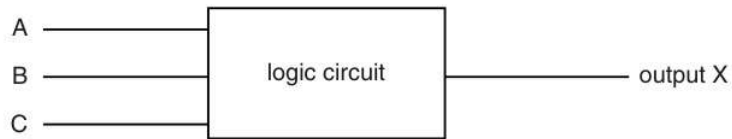
MARK SCHEME

6(a)	1 mark per gate The OR gates may be re-sequenced  Alternatively: 3-input OR gate 1 mark for first three gates, and 2 marks for 3-input OR gate 	5
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QUESTION

- 6 (a) Three digital sensors A, B and C are used to monitor a process. The outputs from the sensors are used as the inputs to a logic circuit.

A signal, X, is output from the logic circuit:



Output, X, has a value of 1 if either of the following two conditions occur:

- sensor A outputs the value 1 OR sensor B outputs the value 0
- sensor B outputs the value 1 AND sensor C outputs the value 0

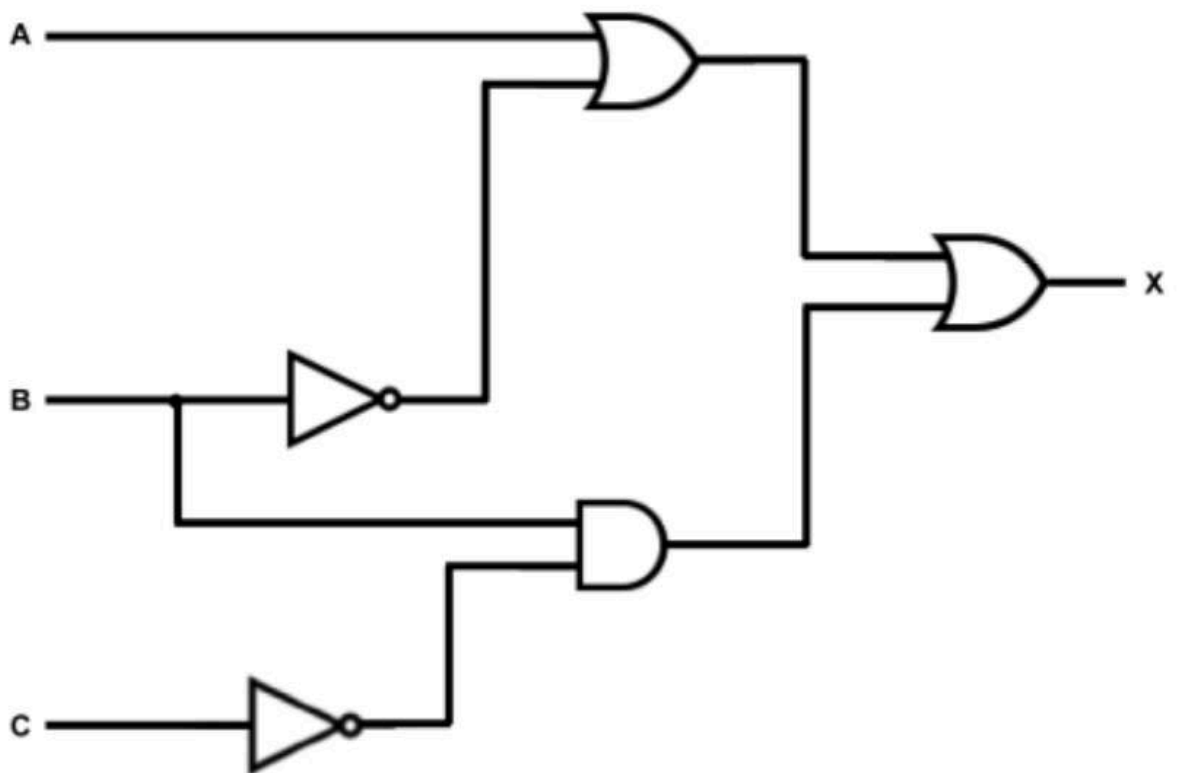
Draw a logic circuit to represent these conditions.



[5]

MARK SCHEME

6 (a)



[5]

QUESTION

- 7 A system is monitored using sensors. The sensors output binary values corresponding to physical conditions, as shown in the table:

Parameter	Description of parameter	Binary value	Description of condition
P	oil pressure	1	pressure $\geq$ 3 bar
		0	pressure $<$ 3 bar
T	temperature	1	temperature $\geq$ 200°C
		0	temperature $<$ 200°C
R	rotation	1	rotation $\leq$ 1000 revs per minute (rpm)
		0	rotation $>$ 1000 revs per minute (rpm)

The outputs of the sensors form the inputs to a logic circuit. The output from the circuit, X, is 1 if any of the following three conditions occur:

- either** oil pressure $\geq$ 3 bar **and** temperature $\geq$ 200°C
or oil pressure $<$ 3 bar **and** rotation $>$ 1000 rpm
or temperature $\geq$ 200°C **and** rotation $>$ 1000 rpm

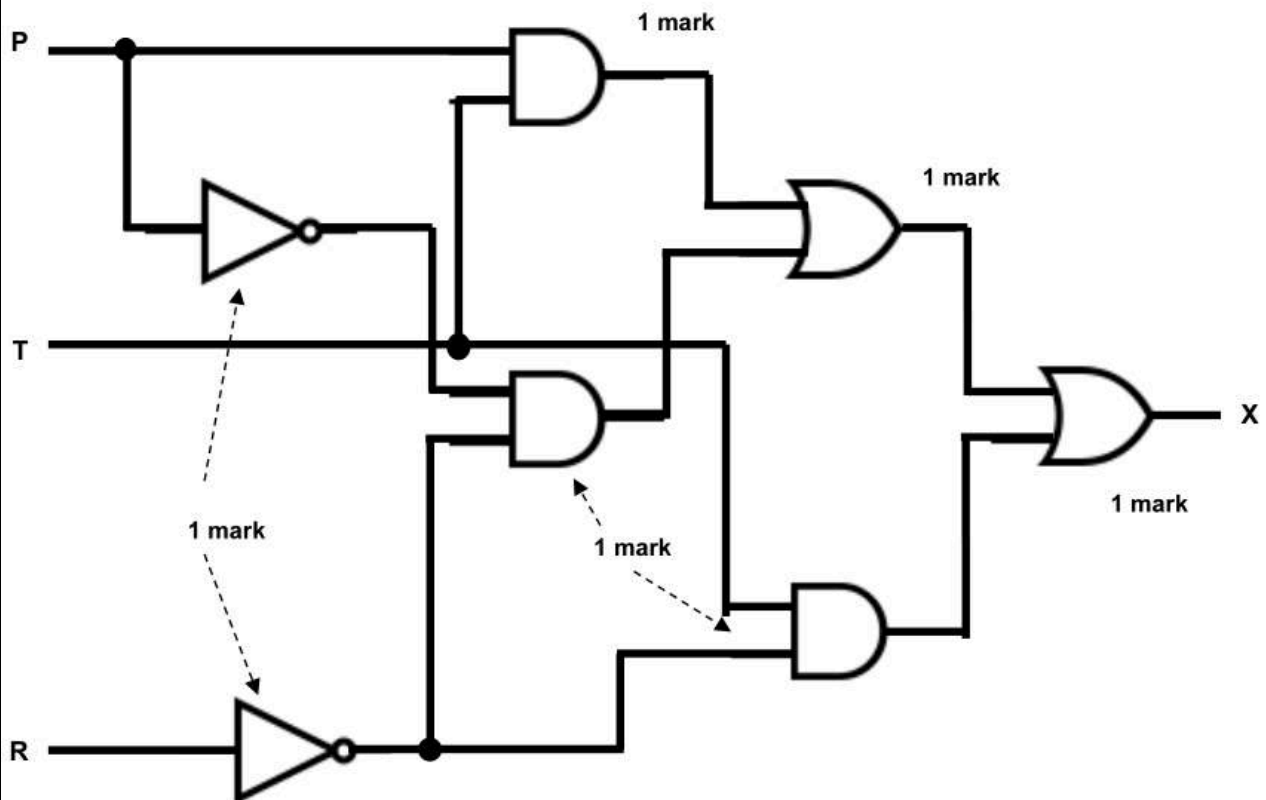
- (a) Draw a logic circuit to represent the above system.



[5]

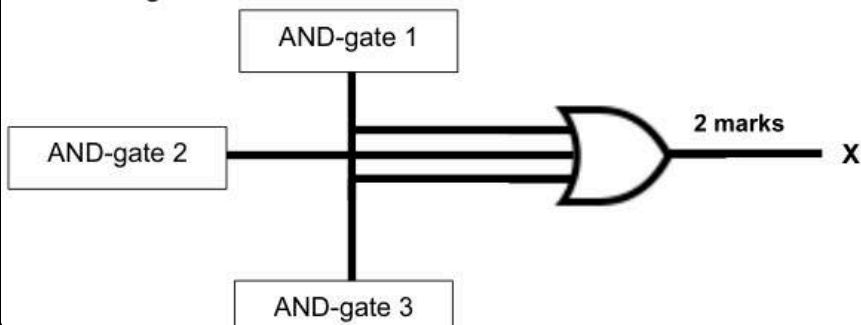
MARK SCHEME

- 7 (a) Since it is possible to simplify the original conditions, at least 3 possible answers exist for the logic circuit.

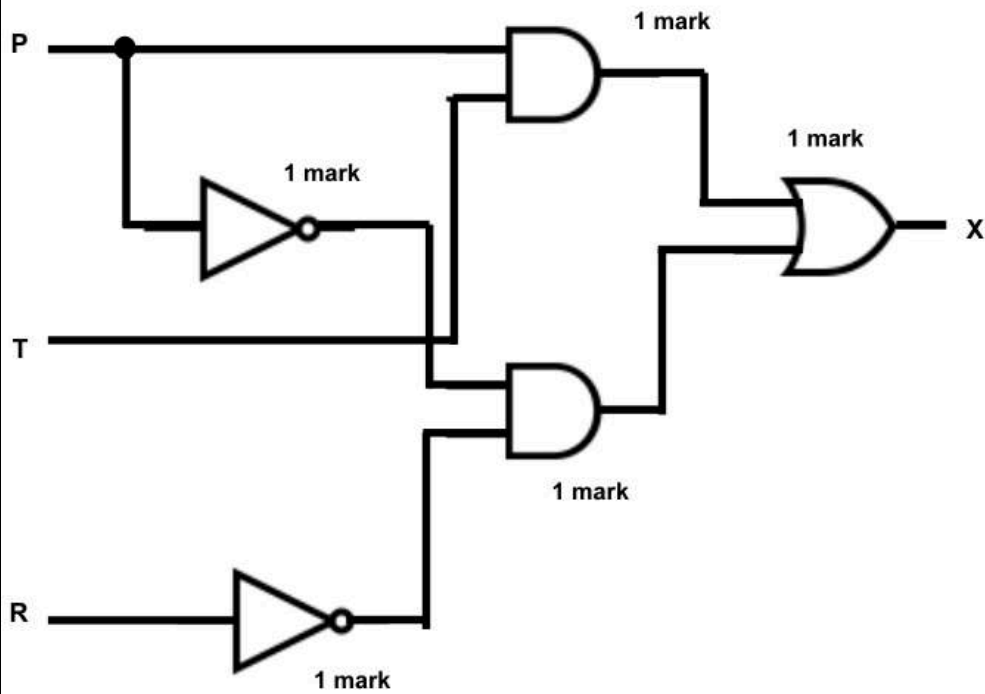


Note: input T has 2 cross overs that should not be connections

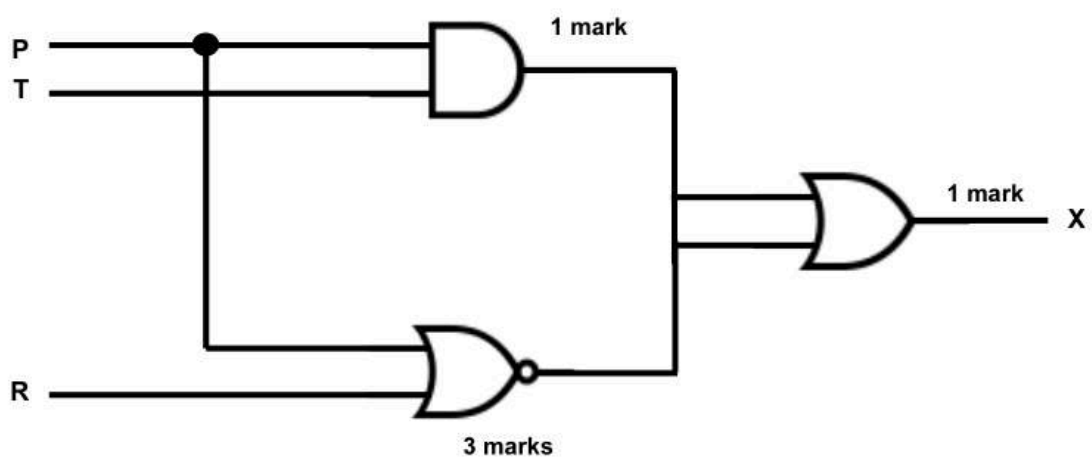
Note: it is possible to use a 3-input OR gate rather than the two 2-input OR gates on the top right:



Alternative solution 1:



Alternative solution 2:



[5]

Note: other solutions may be possible depending on how simplification of the original statement is done

Understand and define the functions of: NOT, AND, OR, NAND, NOR and XOR (EOR) gates

9608_s20_qp_13_sc_3.c

QUESTION

(c) The following is a logic expression.

$$A \text{ AND } B \text{ XOR } C \text{ OR NOT } A$$

Identify **one** logic gate that would **not** be used in the logic circuit for this expression.
Draw the symbol for the logic gate.

Logic gate

Logic gate symbol:

[2]

MARK SCHEME

3(c)	1 mark for name, 1 mark for symbol matching the name	2
NAND		
NOR		

Construct the truth table for each of the logic gates above

9608_s20_qp_11_sc_5.b

QUESTION

(b) Identify **one** logic gate not used in the logic circuit in **part (a)**.

Draw the symbol for this logic gate **and** complete its truth table.

Logic gate:

Symbol:

Truth table:

Input		Output
A	B	
0	0	
0	1	
1	0	
1	1	

[3]

MARK SCHEME

5(b)

1 mark for name, **1 mark** for symbol, **1 mark** for truth table

3

- NAND



Input		Output
A	B	
0	0	1
0	1	1
1	0	1
1	1	0

- NOR



Input		Output
A	B	
0	0	1
0	1	0
1	0	0
1	1	0

Construct a logic circuit

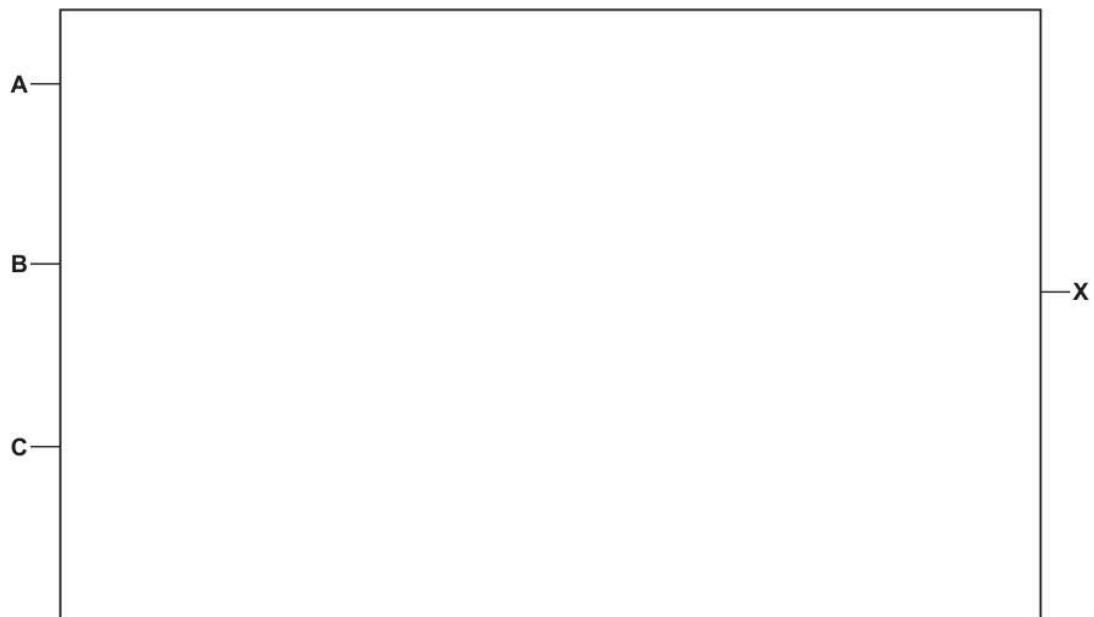
9608_s20_qp_13_sc_3.a

QUESTION

- 3 (a) The following is a logic expression.

$$X = \text{NOT}(A \text{ OR } B) \text{ OR } (A \text{ AND } (B \text{ XOR } C))$$

Draw the logic circuit for the given expression, using a maximum of **four** logic gates.



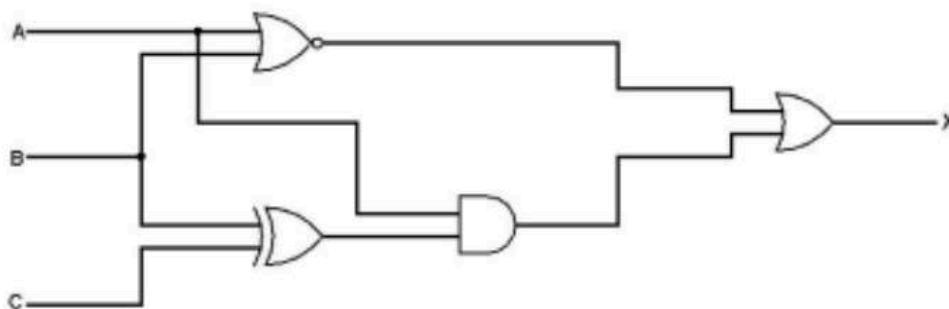
[4]

MARK SCHEME

3(a)

1 mark per correct gate

- A NOR B
- B XOR C
- A AND (B XOR C)
- Final OR



4

QUESTION

- 3 The following is a logic expression.

$$X = \text{NOT } (A \text{ AND } B) \text{ OR NOT } (\text{NOT } B \text{ OR } C)$$

Draw the logic circuit for the given expression using a maximum of **four** gates.

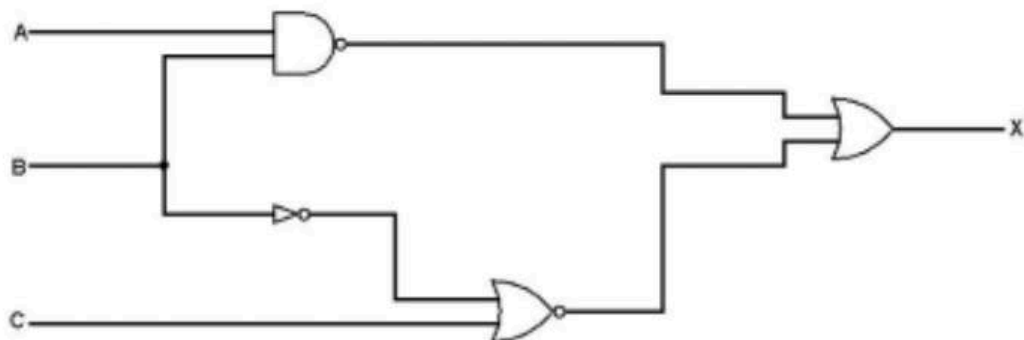


MARK SCHEME

3

1 mark for each correct gate

$$X = \text{NOT } (A \text{ AND } B) \text{ OR NOT } (\text{NOT } B \text{ OR } C)$$



4

QUESTION

- 5 (a) Draw a logic circuit to represent the logic expression:

$$X = \text{NOT } (A \text{ OR } C) \text{ OR } (A \text{ AND NOT } B)$$



[5]

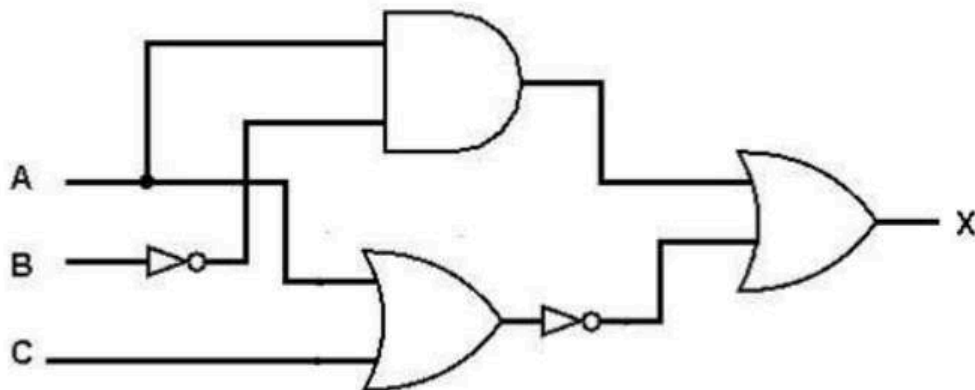
MARK SCHEME

5(a)

1 mark for each correct gate

5

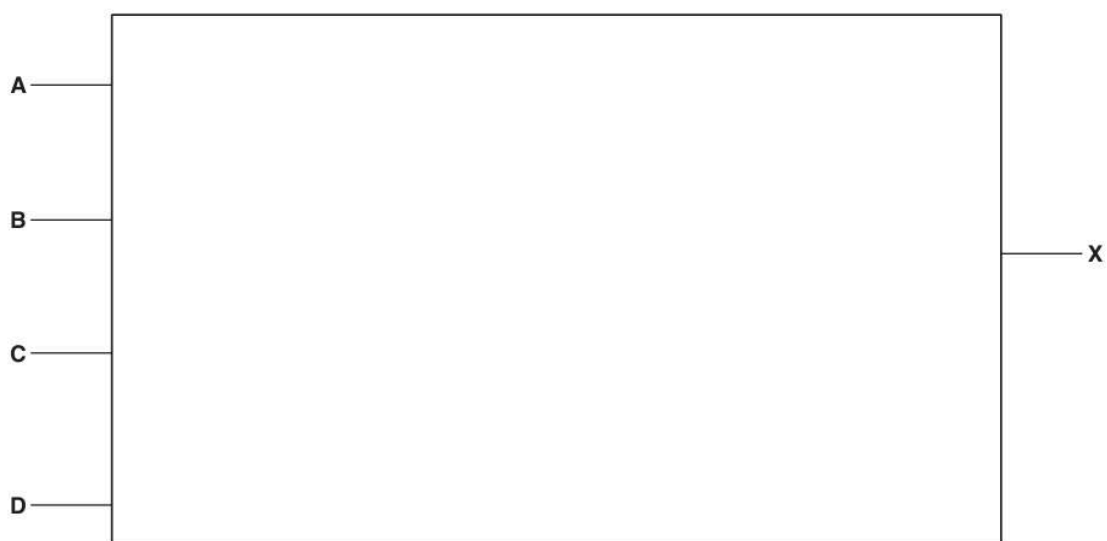
A OR C
NOT(A OR C)
NOT B
A AND NOT B
Final OR



QUESTION

2 (a) Draw a logic circuit to represent the following logic expression:

$$X = \text{NOT } (A \text{ AND } B) \text{ AND } (C \text{ XOR } D)$$



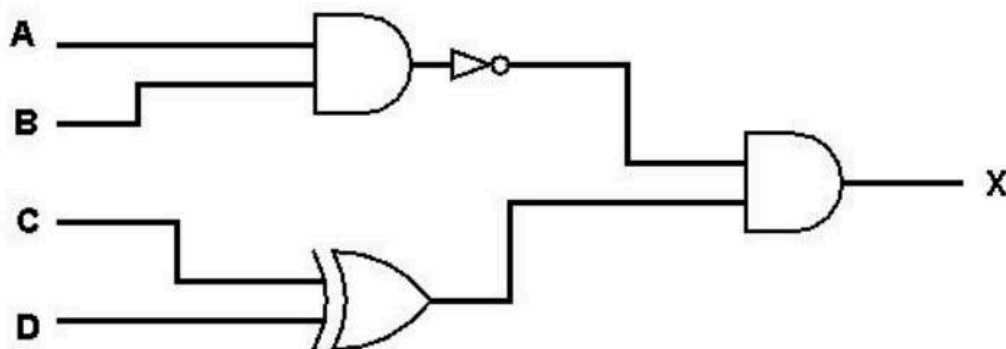
[4]

MARK SCHEME

2(a)

1 mark per gate

4



QUESTION

- 3 (a) A bank approves a customer for an account based on the criteria in the following table.

Parameter	Description of parameter	Binary value	Condition
A	Employed	1	True
		0	False
B	Self-employed	1	True
		0	False
C	Over 21	1	True
		0	False
D	Earn more than 30 000	1	True
		0	False
E	Another account	1	True
		0	False

A customer is approved ($X = 1$) if the person:

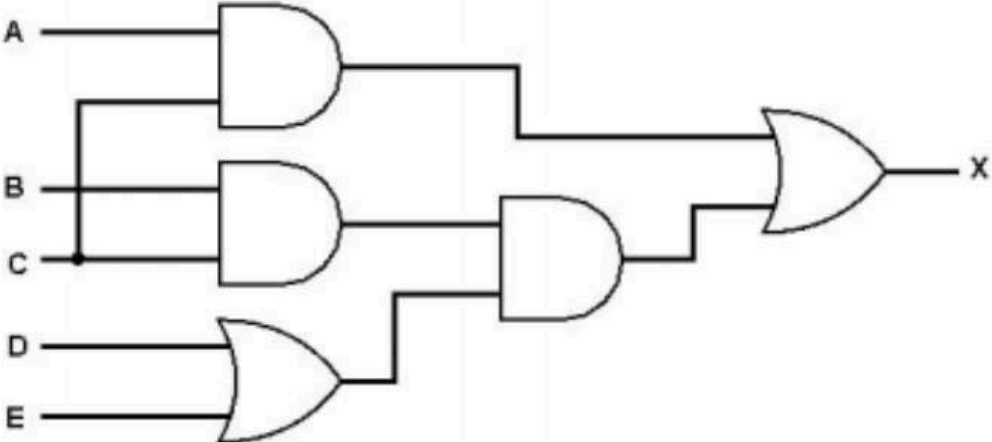
- is over 21 **and** employed
or
- is over 21 **and** self-employed **and**
 - **either** earns more than 30 000
or
 - has another account.

Draw a logic circuit to represent the model.



[5]

MARK SCHEME

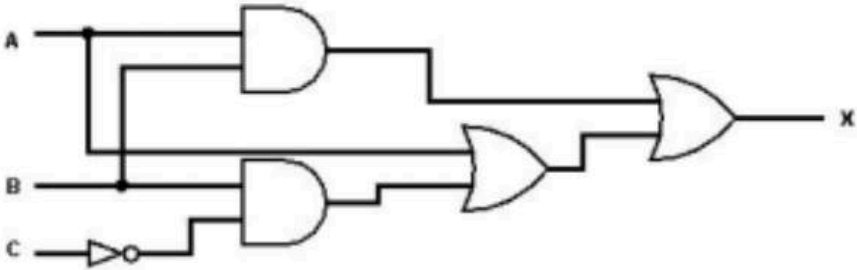
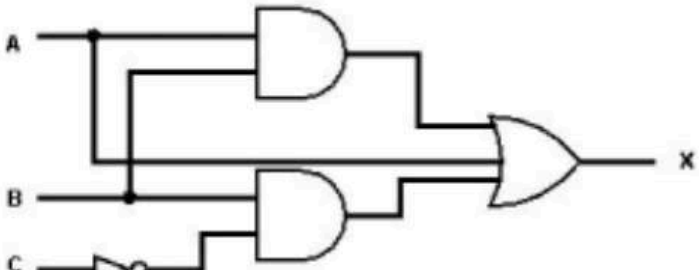
3(a)	1 mark for each gate A AND C B AND C D OR E (B AND C) AND (D OR E) Final OR 	5
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9608_s19_qp_12_sc_6.a

QUESTION

6 (a) Draw a logic circuit to represent the logic expression: $X = A \text{ OR } (B \text{ AND NOT } C) \text{ OR } (A \text{ AND } B)$  [5]

MARK SCHEME

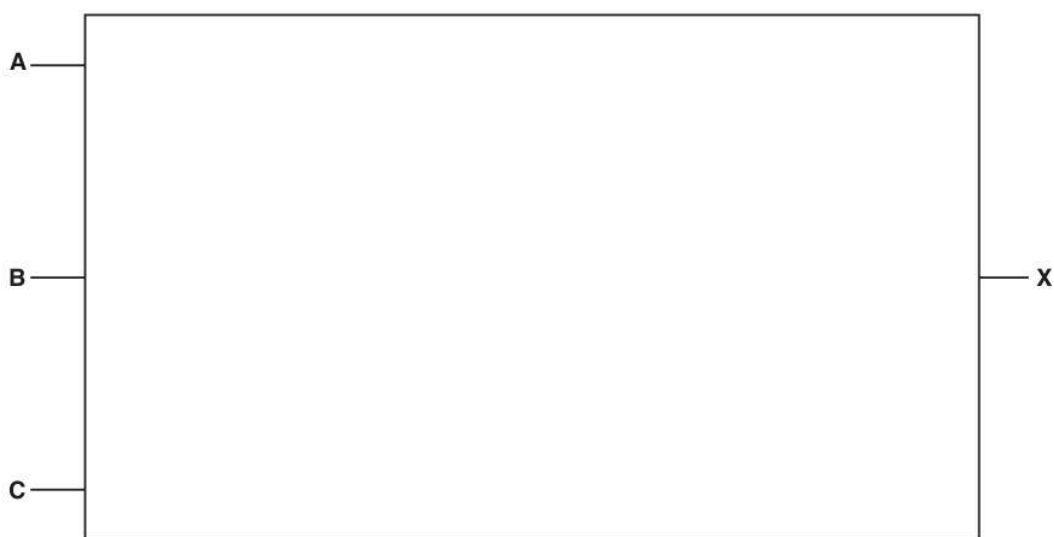
6(a)	1 mark per gate The OR gates may be re-sequenced  Alternatively: 3-input OR gate 1 mark for first three gates, and 2 marks for 3-input OR gate 	5
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9608_w18_qp_13_sc_8.a

QUESTION

- 8 (a)** Draw a logic circuit to represent the logic expression:

$$X = (A \text{ XOR } B) \text{ OR } (\text{NOT}(C \text{ AND } A))$$



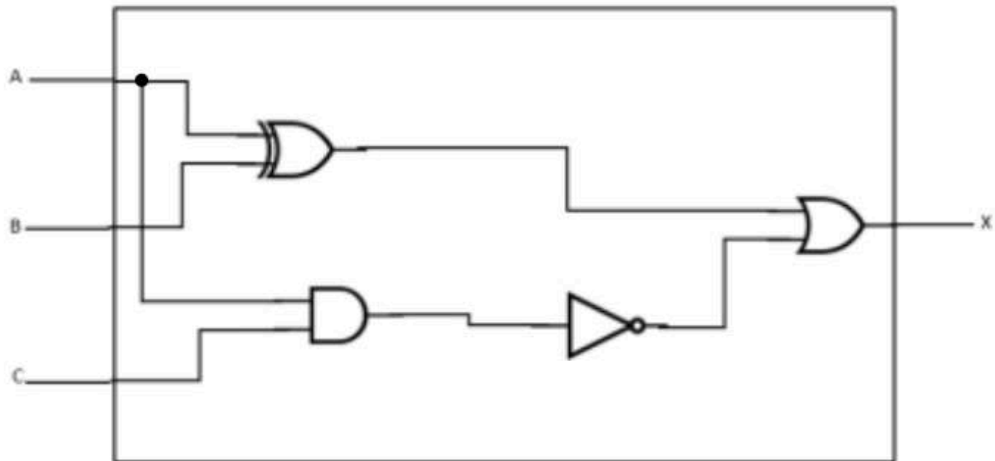
[4]

MARK SCHEME

8(a)

1 mark for each logic gate with the correct inputs

4



QUESTION

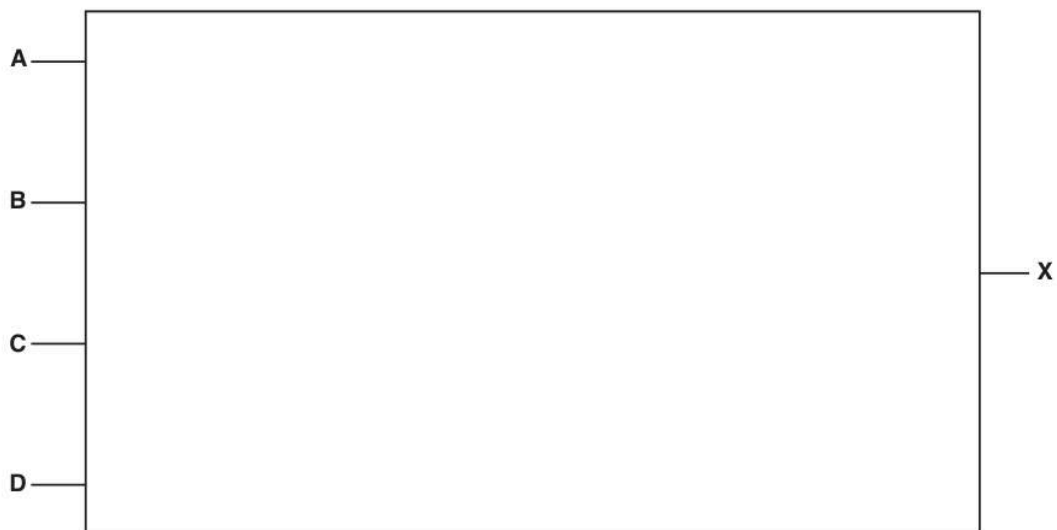
- 5 (a) A student needs to design a logic circuit to model the requirements for membership of a snooker club. Membership (X) depends on four criteria, as shown in the table:

Parameter	Description of parameter	Binary value	Condition
A	Over 18	1	True
		0	False
B	Recommended	1	True
		0	False
C	Full-time	1	True
		0	False
D	Retired	1	True
		0	False

Membership is approved ($X = 1$) if the person:

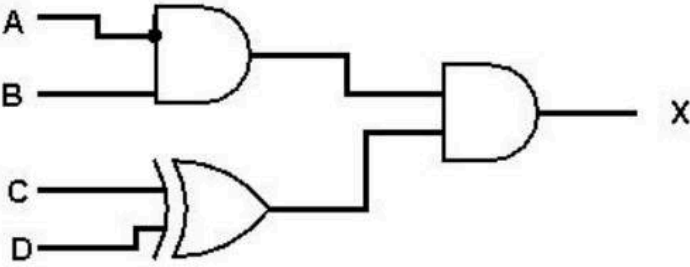
- is over the age of 18 **and** has been recommended by a pre-existing member **and**
- **either** is working full-time **or** is retired, but not both.

Draw a logic circuit to represent the membership requirements.



[3]

MARK SCHEME

5(a)	1 mark per correct gate with correct inputs  <pre data-bbox="316 297 1007 562">graph LR; A --> AND1[AND]; B --> AND1; C --> OR1[OR]; D --> OR1; AND1 --> AND2[AND]; OR1 --> AND2; AND2 --> X[X];</pre>	3
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QUESTION

- 2 (a) A greenhouse control system has four input parameters (H, D, T, W) and two outputs (X, Y).

Parameter	Description of parameter	Binary value	Condition
H	Humidity	0	Too low
		1	Acceptable
D	Day	0	Night
		1	Day
T	Temperature	0	Too high
		1	Acceptable
W	Windows	0	Closed
		1	Open

The watering system turns on ($X = 1$) if:

either it is daytime **and** the temperature is too high

or the humidity is too low.

The fan turns on ($Y = 1$) if the temperature is too high **and** the windows are closed.

Draw a logic circuit to represent the greenhouse control system.



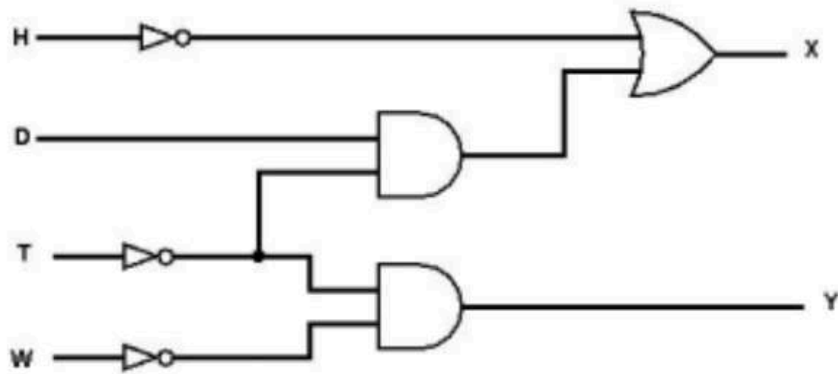
[6]

MARK SCHEME

2(a)

1 mark for each gate with the correct inputs.
Final two gates must also have the correct output.

6



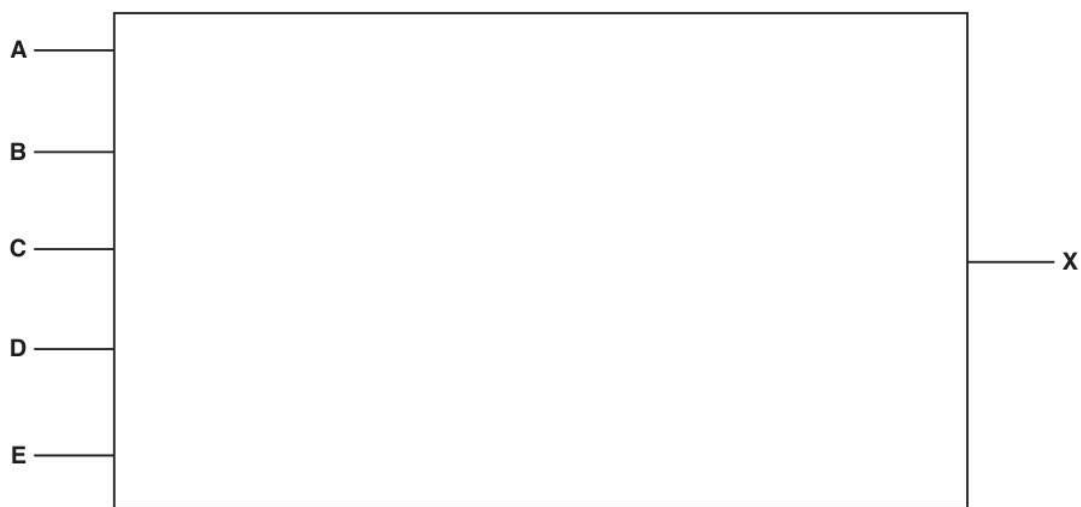
QUESTION

- 4 (a) An alarm system (X) is enabled and disabled using either a switch (A) or a remote control (B). There are **two** infra-red sensors (C, D) and **one** door pressure sensor (E).

Parameter	Description of parameter	Binary value	Condition
A	Switch	1	Switch enabled
		0	Switch disabled
B	Remote control	1	Remote enabled
		0	Remote disabled
C	Infra-red sensor	1	Activated
		0	Not activated
D	Infra-red sensor	1	Activated
		0	Not activated
E	Door pressure sensor	1	Activated
		0	Not activated

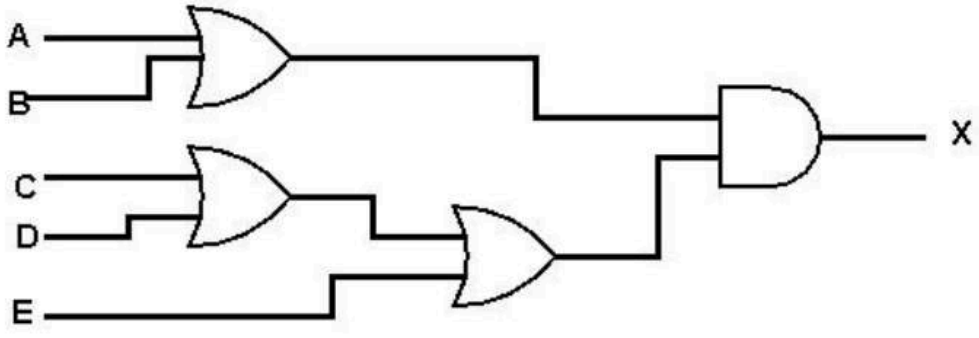
The alarm sounds ($X = 1$) if the alarm is enabled **and** any one or more of the sensors is activated.

Draw a logic circuit to represent the alarm system.



[3]

MARK SCHEME

4(a)	1 mark per bullet: A OR B C OR D OR E Final AND  <pre>graph LR; A --- OR1(()); B --- OR1; OR1 --- AND1(()); C --- OR2(()); D --- OR2; OR2 --- OR3(()); E --- OR3; OR3 --- AND2(()); AND2 --- X[X];</pre>	3
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QUESTION

- 5 A motor is controlled by a logic circuit. The circuit has inputs (0 or 1) from three sensors R, T and W. The motor is switched off when the output from the logic circuit is 1.

The following table shows the three sensors and the conditions being monitored.

Sensor	Description	Binary value	Condition
R	rotation	0	rotation < 4000 rpm
		1	rotation $\geq$ 4000 rpm
T	temperature	0	temperature $\geq$ 90 °C
		1	temperature < 90 °C
W	water flow rate	0	water flow rate $\geq$ 50 litre/min
		1	water flow rate < 50 litre/min

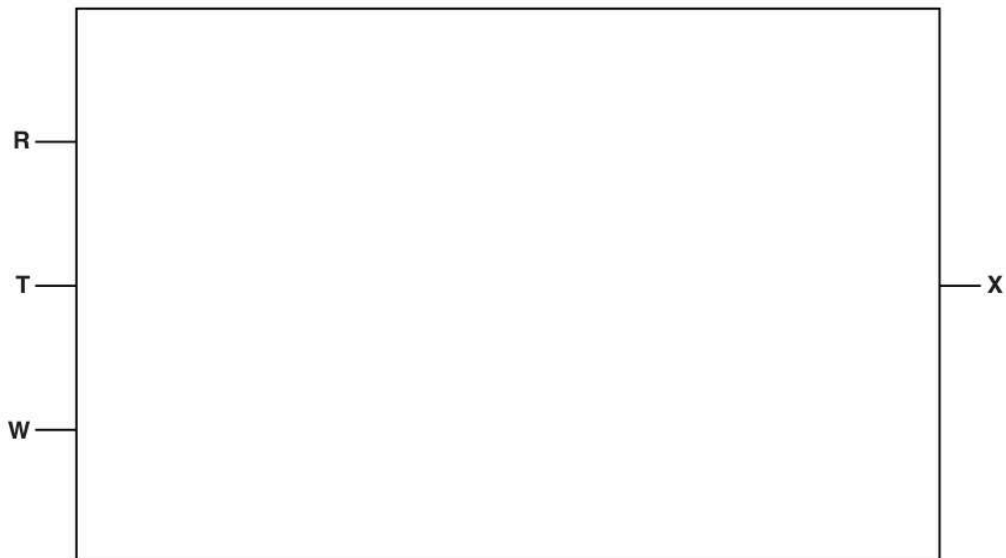
The output, X, is 1 if:

temperature $\geq$ 90 °C and rotation $\geq$ 4000 rpm

or

temperature < 90 °C and water flow rate $\geq$ 50 litre/min

- (i) Draw a corresponding logic circuit.



[5]

MARK SCHEME

9608_w16_qp_12_sc_1.a

QUESTION

- 1 (a) A student writes the following logic expression:

X is 1 IF (B is NOT 1 AND S is NOT 1) OR (P is NOT 1 AND S is 1)

Draw a logic circuit to represent this logic expression.

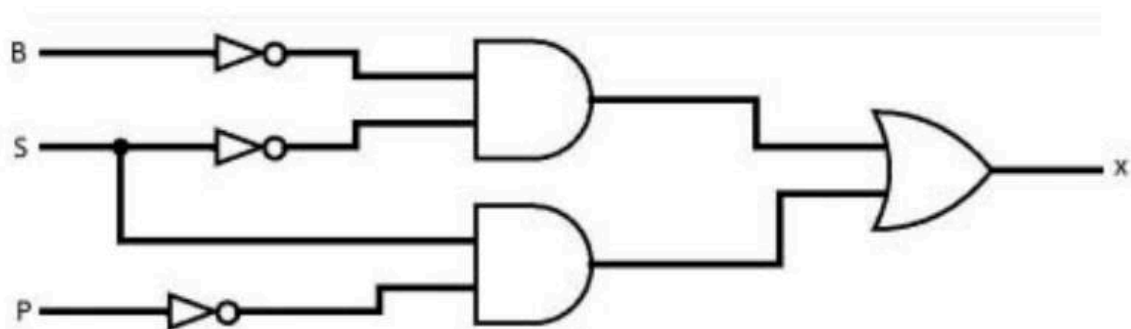
Do not attempt to simplify the logic expression.



[6]

MARK SCHEME

- 1 (a) ONE mark for each correct gate.



[6]

QUESTION

- 5 A motor is controlled by a logic circuit. The circuit has inputs (0 or 1) from three sensors R, T and W. The motor is switched off when the output from the logic circuit is 1.

The following table shows the three sensors and the conditions being monitored.

Sensor	Description	Binary value	Condition
R	rotation	0	rotation < 4000 rpm
		1	rotation $\geq$ 4000 rpm
T	temperature	0	temperature $\geq$ 90 °C
		1	temperature < 90 °C
W	water flow rate	0	water flow rate $\geq$ 50 litre/min
		1	water flow rate < 50 litre/min

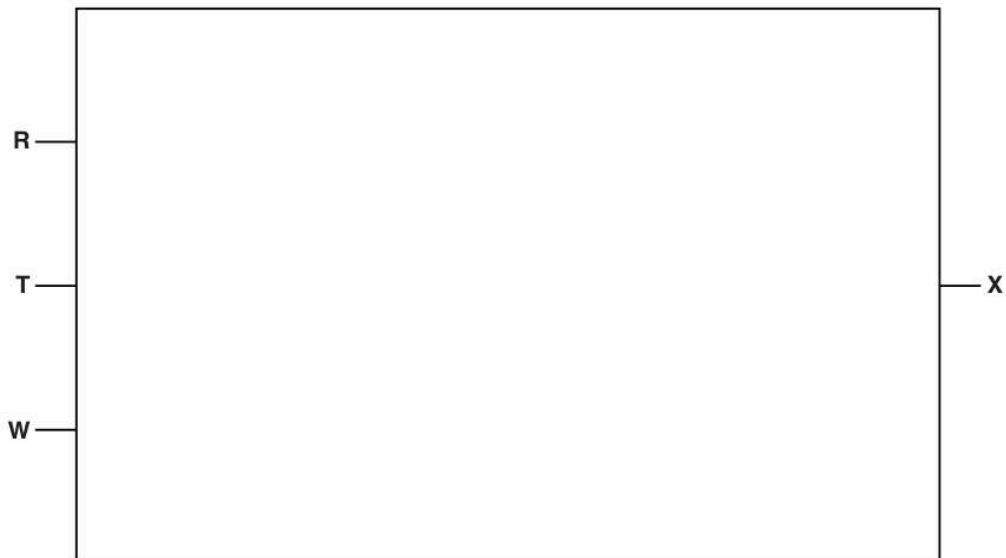
The output, X, is 1 if:

temperature $\geq$ 90 °C and rotation $\geq$ 4000 rpm

or

temperature < 90 °C and water flow rate $\geq$ 50 litre/min

- (i) Draw a corresponding logic circuit.



[5]

QUESTION

- 6 (a) Three digital sensors A, B and C are used to monitor a process. The outputs from the sensors are used as the inputs to a logic circuit.

A signal, X, is output from the logic circuit:



Output, X, has a value of 1 if either of the following two conditions occur:

- sensor A outputs the value 1 OR sensor B outputs the value 0
- sensor B outputs the value 1 AND sensor C outputs the value 0

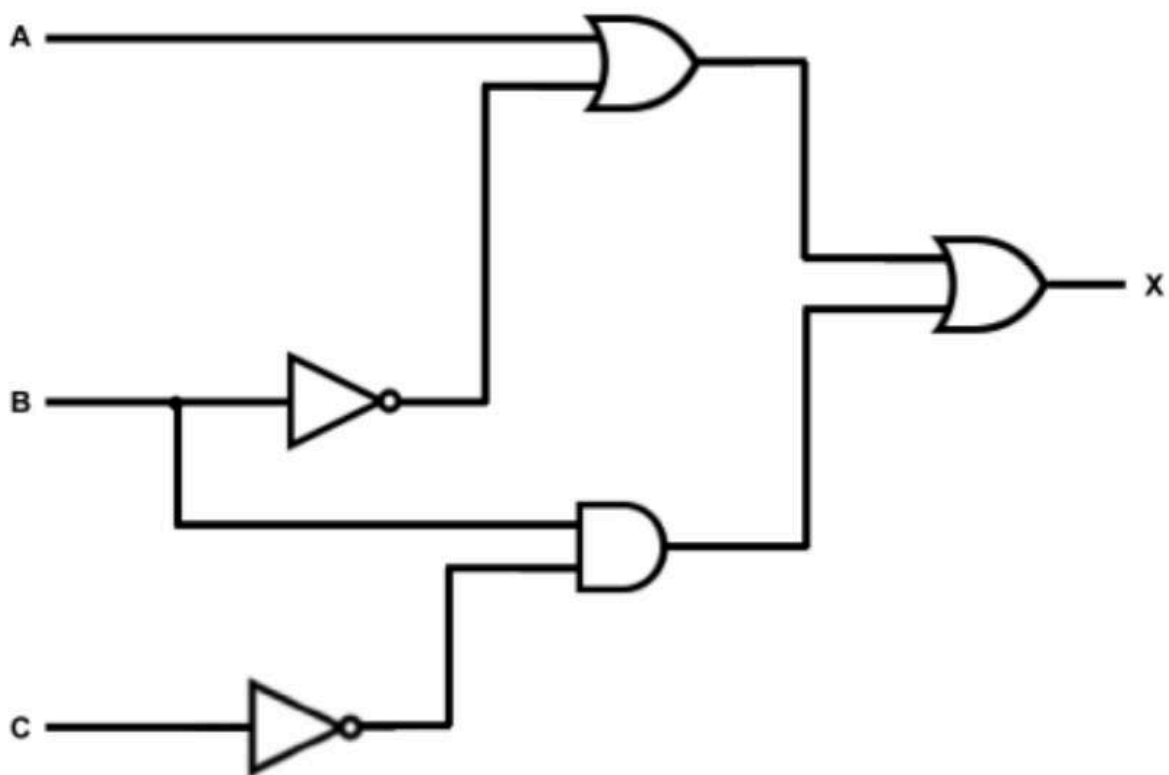
Draw a logic circuit to represent these conditions.



[5]

MARK SCHEME

6 (a)



[5]

QUESTION

- 7 A system is monitored using sensors. The sensors output binary values corresponding to physical conditions, as shown in the table:

Parameter	Description of parameter	Binary value	Description of condition
P	oil pressure	1	pressure $\geq$ 3 bar
		0	pressure $<$ 3 bar
T	temperature	1	temperature $\geq$ 200°C
		0	temperature $<$ 200°C
R	rotation	1	rotation $\leq$ 1000 revs per minute (rpm)
		0	rotation $>$ 1000 revs per minute (rpm)

The outputs of the sensors form the inputs to a logic circuit. The output from the circuit, X, is 1 if any of the following three conditions occur:

either oil pressure $\geq$ 3 bar **and** temperature $\geq$ 200°C

or oil pressure $<$ 3 bar **and** rotation $>$ 1000 rpm

or temperature $\geq$ 200°C **and** rotation $>$ 1000 rpm

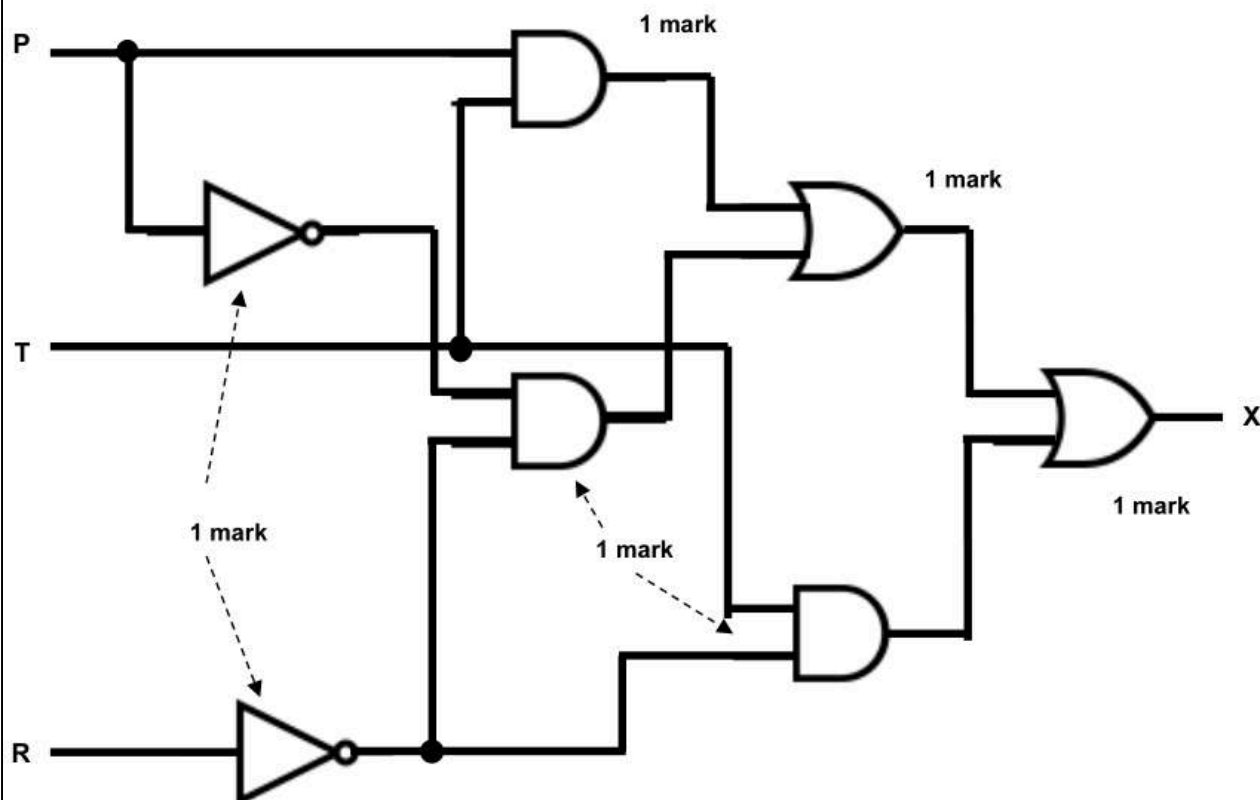
- (a) Draw a logic circuit to represent the above system.



[5]

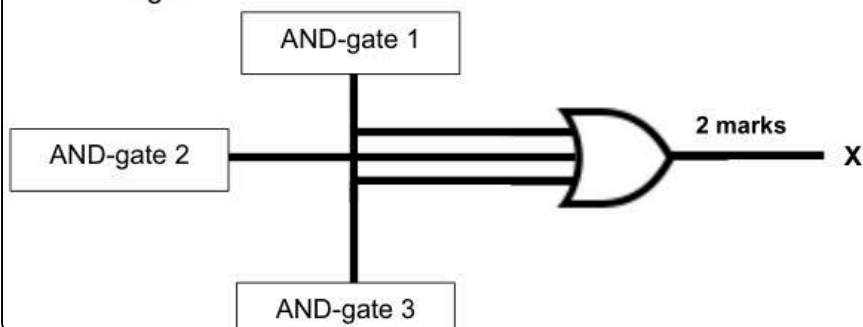
MARK SCHEME

- 7 (a) Since it is possible to simplify the original conditions, at least 3 possible answers exist for the logic circuit.

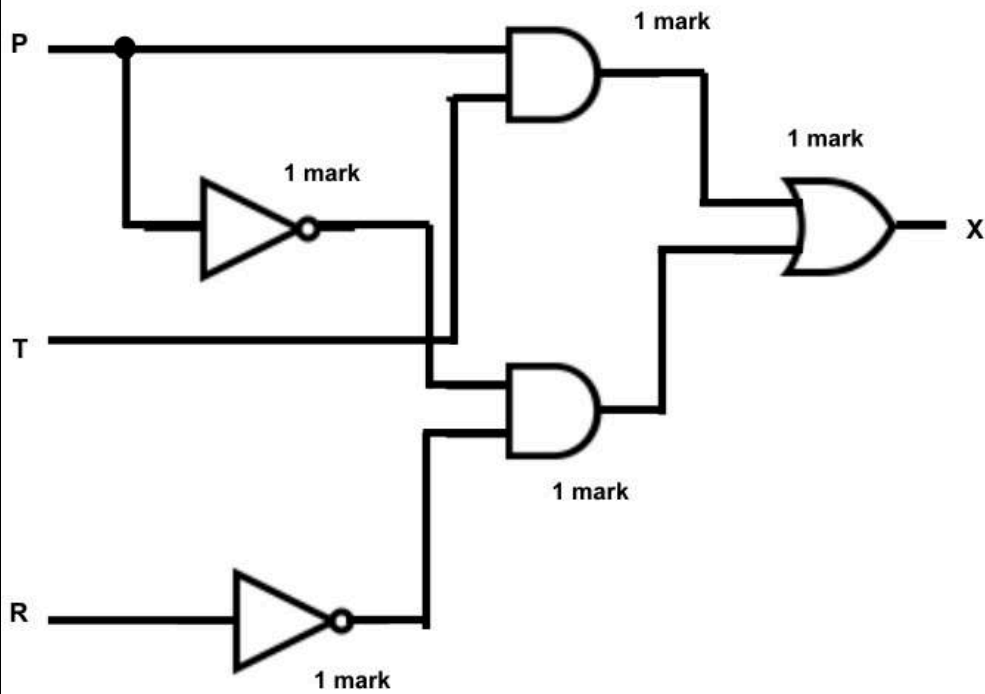


Note: input T has 2 cross overs that should not be connections

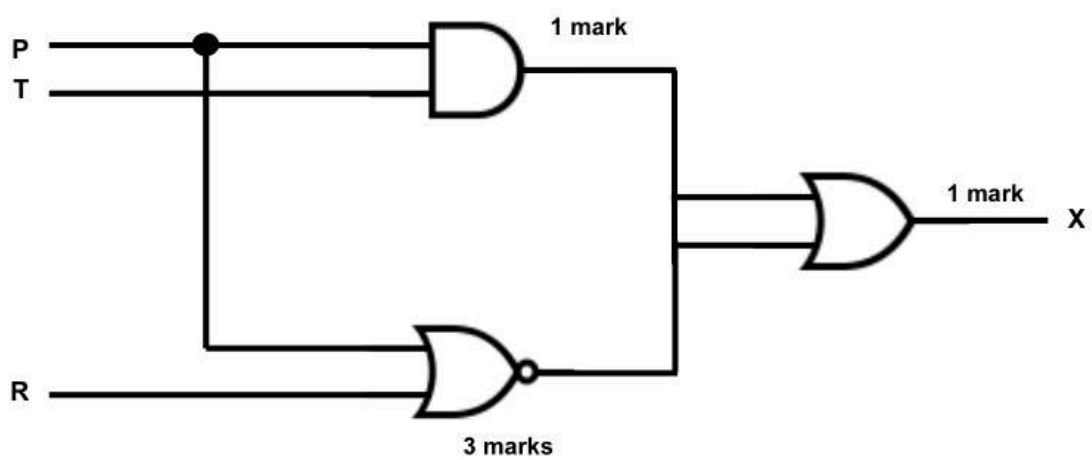
Note: it is possible to use a 3-input OR gate rather than the two 2-input OR gates on the top right:



Alternative solution 1:



Alternative solution 2:



[5]

Note: other solutions may be possible depending on how simplification of the original statement is done

QUESTION

- 7 A system is monitored using sensors. The sensors output binary values corresponding to physical conditions, as shown in the table:

Parameter	Description of parameter	Binary value	Description of condition
P	oil pressure	1	pressure $\geq$ 3 bar
		0	pressure $<$ 3 bar
T	temperature	1	temperature $\geq$ 200°C
		0	temperature $<$ 200°C
R	rotation	1	rotation $\leq$ 1000 revs per minute (rpm)
		0	rotation $>$ 1000 revs per minute (rpm)

The outputs of the sensors form the inputs to a logic circuit. The output from the circuit, X, is 1 if any of the following three conditions occur:

either oil pressure $\geq$ 3 bar **and** temperature $\geq$ 200°C

or oil pressure $<$ 3 bar **and** rotation $>$ 1000 rpm

or temperature $\geq$ 200°C **and** rotation $>$ 1000 rpm

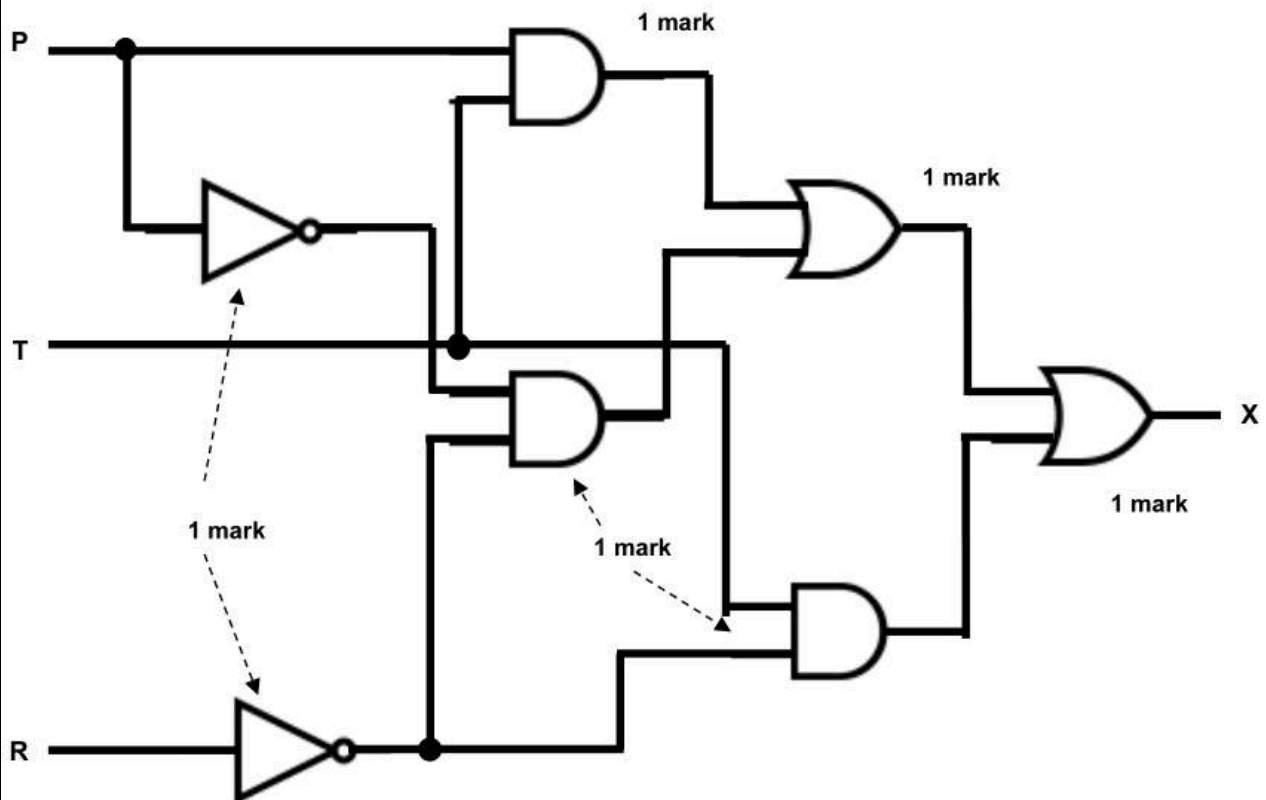
- (a) Draw a logic circuit to represent the above system.



[5]

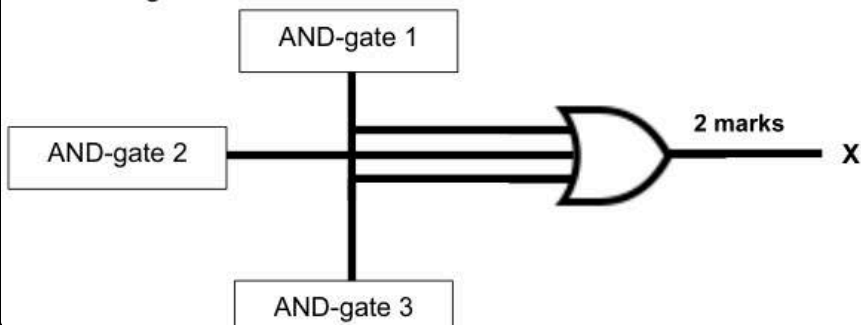
MARK SCHEME

- 7 (a) Since it is possible to simplify the original conditions, at least 3 possible answers exist for the logic circuit.

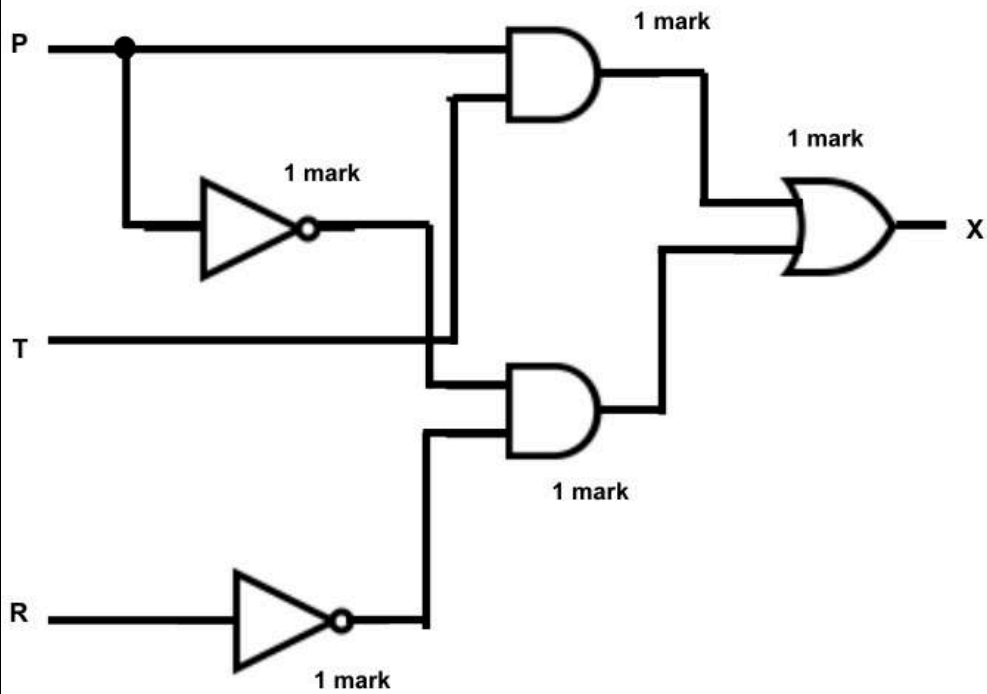


Note: input T has 2 cross overs that should not be connections

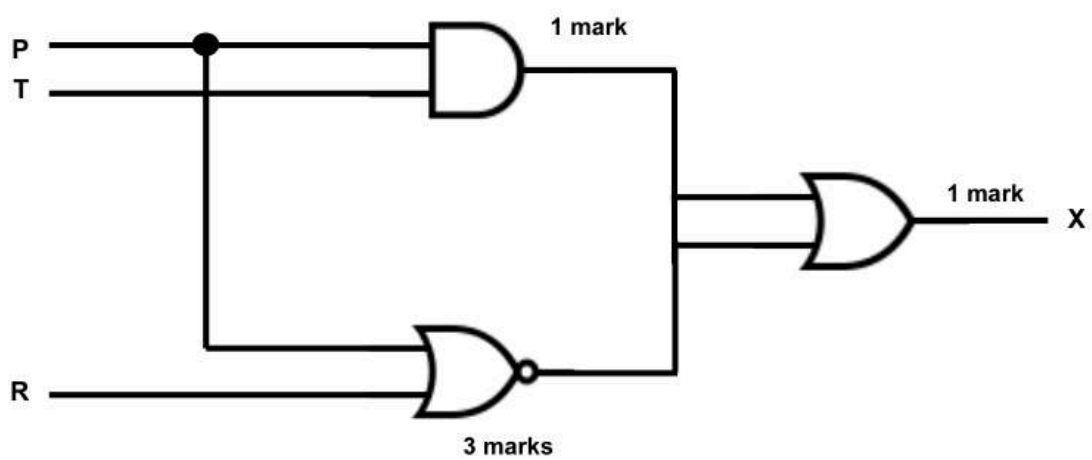
Note: it is possible to use a 3-input OR gate rather than the two 2-input OR gates on the top right:



Alternative solution 1:



Alternative solution 2:



[5]

Note: other solutions may be possible depending on how simplification of the original statement is done

Construct a truth table

9608_w21_qp_11_sc_5

QUESTION

- 5 Complete the truth table for the following logic expression:

$$X = \text{NOT}(A \text{ OR } B) \text{ AND NOT}(\text{NOT}(B \text{ OR } C) \text{ AND } (\text{NOT } A))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

MARK SCHEME

5

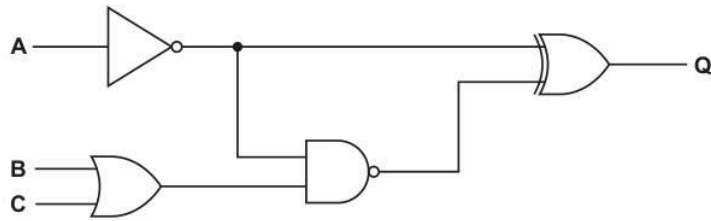
One mark per pair of rows (shaded and unshaded)

4

A	B	C	Working space	X
0	0	0		0
0	0	1		1
0	1	0		0
0	1	1		0
1	0	0		0
1	0	1		0
1	1	0		0
1	1	1		0

QUESTION

2 Consider the following logic circuit:



(a) Complete the truth table for the logic circuit.

A	B	C	Working space	Q
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

MARK SCHEME

2(a)

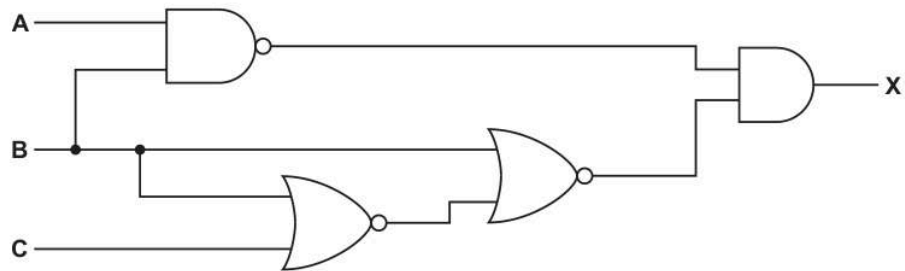
1 mark per pair of outputs (shaded)

4

A	B	C	Working space	Q
0	0	0		0
0	0	1		1
0	1	0		1
0	1	1		1
1	0	0		1
1	0	1		1
1	1	0		1
1	1	1		1

QUESTION

4 Consider the following logic circuit:



(a) Complete the truth table for the logic circuit.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

(b) Identify three logic gates not used in the logic circuit.

MARK SCHEME

4(a)

1 mark per pair of outputs (shaded)

4

A	B	C	Working space	X
0	0	0		0
0	0	1		1
0	1	0		0
0	1	1		0
1	0	0		0
1	0	1		1
1	1	0		0
1	1	1		0

9608_s20_qp_13_sc_3.b

QUESTION

(b) Complete the truth table for the logic expression:

$$X = \text{NOT}(A \text{ OR } B) \text{ OR } (A \text{ AND } (B \text{ XOR } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

MARK SCHEME

3(b)

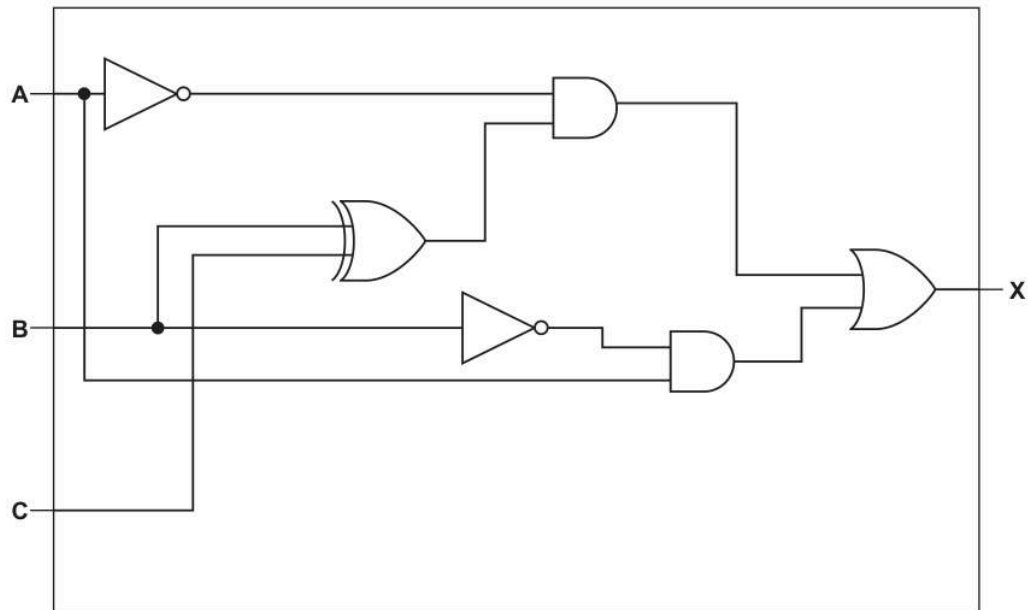
1 mark for each correct pair of answers (4 shaded sections)

4

A	B	C	Working Space	X
0	0	0		1
0	0	1		1
0	1	0		0
0	1	1		0
1	0	0		0
1	0	1		1
1	1	0		1
1	1	1		0

QUESTION

5 (a) A logic circuit is given:



Complete the following truth table for the logic circuit.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

MARK SCHEME

5(a)

1 mark for each pair of correct answers (shaded)

4

A	B	C	Working space	X
0	0	0		0
0	0	1		1
0	1	0		1
0	1	1		0
1	0	0		1
1	0	1		1
1	1	0		0
1	1	1		0

9608_w19_qp_13_sc_5.b

QUESTION

(b) Complete the truth table for the logic expression:

$$X = \text{NOT } (A \text{ OR } C) \text{ OR } (A \text{ AND NOT } B)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

MARK SCHEME

5(b)

1 mark for each pair of rows

4

A	B	C	Working space	X
0	0	0		1
0	0	1		0
0	1	0		1
0	1	1		0
1	0	0		1
1	0	1		1
1	1	0		0
1	1	1		0

9608_w19_qp_12_sc_2.b

QUESTION

(b) Complete the truth table for the logic expression:

$$X = \text{NOT } (A \text{ AND } B) \text{ OR } (A \text{ AND } (B \text{ XOR } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

MARK SCHEME

2(b)	1 mark for each pair of answers	4
------	---------------------------------	---

A	B	C	Working space	X
0	0	0		1
0	0	1		1
0	1	0		1
0	1	1		1
1	0	0		1
1	0	1		1
1	1	0		1
1	1	1		0

9608_w19_qp_11_sc_3.b

QUESTION

(b) Complete the truth table for the logic expression:

$$X = (A \text{ AND } C) \text{ OR } (\text{NOT } A \text{ AND } (B \text{ XOR } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

MARK SCHEME

3(b)

1 mark for each pair of correct rows

4

A	B	C	Working space	X
0	0	0		0
0	0	1		1
0	1	0		1
0	1	1		0
1	0	0		0
1	0	1		1
1	1	0		0
1	1	1		1

9608_s19_qp_12_sc_6.b

QUESTION

(b) Complete the truth table for the logic expression in **part (a)**.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

MARK SCHEME

6(b)

1 mark for each pair of rows (shaded)

4

A	B	C	Working Space	X
0	0	0		0
0	0	1		0
0	1	0		1
0	1	1		0
1	0	0		1
1	0	1		1
1	1	0		1
1	1	1		1

9608_w18_qp_13_sc_8.b

QUESTION

(b) Complete the truth table for the logic expression in part (a).

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

MARK SCHEME

8(b)

One mark for each correct pair of lines

4

A	B	C	Working Space	X
0	0	0		1
0	0	1		1
0	1	0		1
0	1	1		1
1	0	0		1
1	0	1		1
1	1	0		1
1	1	1		0

9608_s18_qp_13_sc_5.b

QUESTION

(b) Complete the truth table for the logic expression: $X = (A \text{ XOR } B) \text{ AND NOT } C$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

MARK SCHEME

5(b)

1 mark for each correct pair of lines

4

A	B	C	Working space	X
0	0	0		0
0	0	1		0
0	1	0		1
0	1	1		0
1	0	0		1
1	0	1		0
1	1	0		0
1	1	1		0

9608_s18_qp_12_sc_2.b

QUESTION

(b) Complete the truth table for the logic expression: $X = \text{NOT } A \text{ AND } (B \text{ NAND } C)$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

MARK SCHEME

2(b)

One mark for each pair of rows.

4

A	B	C	Working space	X
0	0	0		1
0	0	1		1
0	1	0		1
0	1	1		0
1	0	0		0
1	0	1		0
1	1	0		0
1	1	1		0

9608_s18_qp_11_sc_4.b

QUESTION

(b) Complete the truth table for the logic expression: $X = A \text{ OR } (B \text{ XOR } C)$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

MARK SCHEME

4(b)

1 mark for each correct pair of rows

4

A	B	C	Working space	X
0	0	0		0
0	0	1		1
0	1	0		1
0	1	1		0
1	0	0		1
1	0	1		1
1	1	0		1
1	1	1		1

9608_w16_qp_13_sc_5.iii

QUESTION

(iii) Complete the truth table for this system.

INPUT			Workspace	OUTPUT X
R	T	W		
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

MARK SCHEME

9608_w16_qp_12_sc_1.b

QUESTION

(b) Complete the truth table for the logic expression given in **part (a)**.

B	S	P	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

MARK SCHEME

(b) **ONE** mark for each pair of rows.

B	S	P	Working space	X
0	0	0		1
0	0	1		1
0	1	0		1
0	1	1		0
1	0	0		0
1	0	1		0
1	1	0		1
1	1	1		0

[4]

QUESTION

(iii) Complete the truth table for this system.

INPUT			Workspace	OUTPUT X
R	T	W		
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

MARK SCHEME

QUESTION

(b) Complete the truth table for the logic circuit described in **part (a)**.

A	B	C	Working Space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

MARK SCHEME

(b)

A	B	C	working	X	
0	0	0		1	} 1 mark
0	0	1		1	
0	1	0		1	} 1 mark
0	1	1		0	
1	0	0		1	} 1 mark
1	0	1		1	
1	1	0		1	} 1 mark
1	1	1		1	

[4]

9608_s15_qp_12_sc_7.b

QUESTION

(b) Complete the truth table for this system.

P	T	R	Workspace	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

MARK SCHEME

(b)

P	T	R	Workspace	X
0	0	0		1
0	0	1		0
0	1	0		1
0	1	1		0
1	0	0		0
1	0	1		0
1	1	0		1
1	1	1		1

} 1 mark

} 1 mark

} 1 mark

} 1 mark

[4]

9608_s15_qp_11_sc_7.b

QUESTION

(b) Complete the truth table for this system.

P	T	R	Workspace	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

MARK SCHEME

(b)

P	T	R	Workspace	X
0	0	0		1
0	0	1		0
0	1	0		1
0	1	1		0
1	0	0		0
1	0	1		0
1	1	0		1
1	1	1		1

} 1 mark

} 1 mark

} 1 mark

} 1 mark

[4]

Construct a logic expression

9608_w16_qp_13_sc_5.ii

QUESTION

(ii) Give a logic statement corresponding to the logic circuit in part (i).

.....
.....[2]

MARK SCHEME

9608_w16_qp_11_sc_5.ii

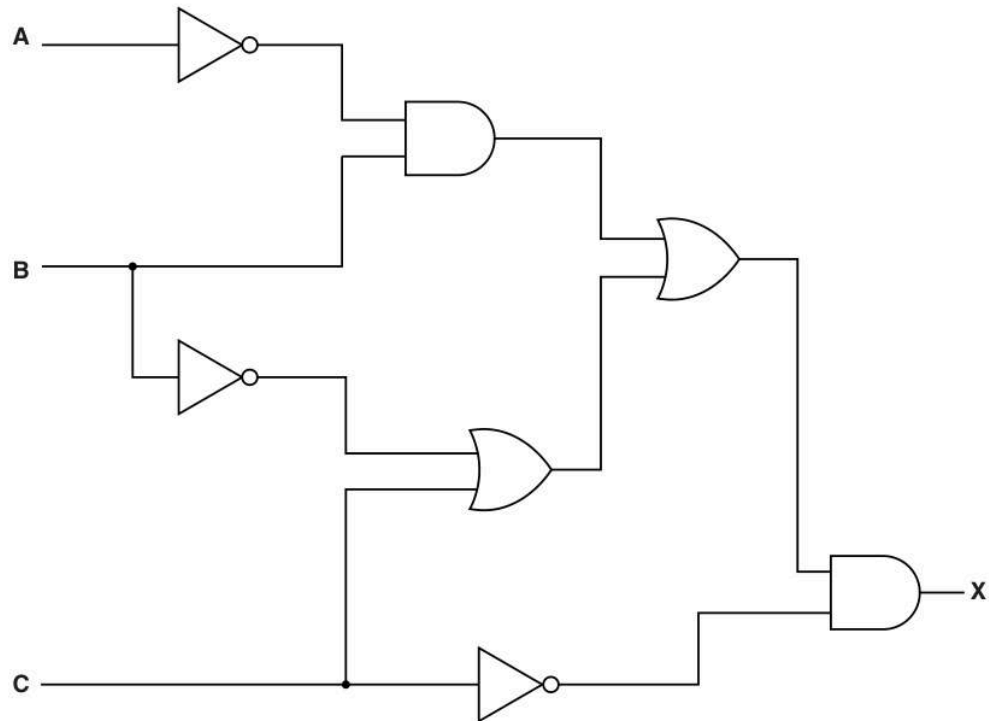
QUESTION

(ii) Give a logic statement corresponding to the logic circuit in part (i).

.....
.....[2]

QUESTION

(c) Write a logic statement that describes the following logic circuit.



.....

 [3]

MARK SCHEME

(c) ((A is NOT 1 AND B is 1) OR (B is NOT 1 OR C is 1)) AND C is NOT 1
<----- 1 mark -----> <----- 1 mark -----> <-----1 mark ----->

NOTE: all brackets may not be shown – but check answer still correct

Alternatives include:

((NOT A AND B) OR (NOT B OR C)) AND NOT C

$(\bar{A} \cdot B + (\bar{B} + C)) \cdot \bar{C}$

NOTE: expressions may be reversed but still OK

(e.g. NOT C AND ((NOT A AND B) OR (NOT B OR C))

NOT C AND ((NOT B OR C) OR (NOT A AND B)) and so on)

[3]